

Sepsis and Suspected Sepsis- Children (Presenting to Child Health and the Emergency Department) Management Guideline and Care Pathway

V2.0

January 2024

Summary

See form at appendix 3.

1. Aim/ Purpose of this Guideline

- 1.1. Clinical Guideline for the management of sepsis in children and young people applies to all those who assess and treat children and young people.
- 1.2. Sepsis is a clinical syndrome complicating infection and resulting in systemic inflammation, microcirculatory disturbance, and end organ dysfunction. Untreated it may progress to septic shock and circulatory collapse.
- 1.3. This version supersedes any previous versions of this document.
- 1.4. Glossary:

- NICE: National Institute for Health and Care Excellence.
- RCHT: Royal Cornwall Hospitals NHS Trust.
- IV: Intravenous.
- IO: Intraosseous.
- ST4: Specialty Trainee Year 4.
- BPM: Beats per minute.
- RPM: Respirations per minutes.
- mmHg: Millimeters of mercury.
- Mls: Milliliters.
- Kg: Kilograms.
- FBC: Full blood count.
- U&Es: Urea and electrolytes.
- Mmol/L: millimoles per litre.
- Ca: Calcium.
- HAS: Human Albumin Solution.
- CRP: C-reactive protein.
- fiO2: Fraction of inspired oxygen.
- Sats: Saturations.
- HR: Heart rate.
- pH: Potential of hydrogen.

- RR: Respiratory rate.
- BE: Barium enema.
- HCO3: Bicarbonate.
- CO2: Carbon dioxide.
- O2: Oxygen.
- SaO2: Oxygen saturation.
- Cap: Capillary.
- Art: Arterial.
- UO: Urinary output.
- CRT: Capillary refill time.
- ITU: Intensive Therapy Unit.
- HDU: High Dependency Unit.
- AKI: Acute kidney injury.
- DKA: Diabetic ketoacidosis.
- Abx: Antibiotics.

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The Trust has a duty under the Data Protection Act 2018 and UK General Data Protection Regulations 2016/679 to ensure that there is a valid legal basis to process personal and sensitive data. The legal basis for processing must be identified and documented before the processing begins. In many cases we may need consent; this must be explicit, informed, and documented. We cannot rely on opt out, it must be opt in.

Data Protection Act 2018 and UK General Data Protection Regulations 2016/679 is applicable to all staff; this includes those working as contractors and providers of services.

For more information about your obligations under the Data Protection Act 2018 and UK General Data Protection Regulations 2016/679 please see the Information Use Framework Policy or contact the Information Governance Team.

Royal Cornwall Hospital Trust rch-tr.infogov@nhs.net

2. The Guidance

2.1. General points for consideration:

- 2.1.1. Sepsis should be considered in all unwell children and initiating treatment early and if suspected clinically treatment should usually be initiated without delay.
- 2.1.2. Be aware that children and young people with sepsis may have non-specific, non-localising presentations (for example, feeling very unwell / reduced activity/ lethargy).
- 2.1.3. Pay particular attention to concerns expressed by the young person and family/ carer and representations with the same illness.
- 2.1.4. Take particular care in the assessment of children and young people with communication difficulties and or complex needs.

2.2. Risk factors:

The following groups of children and young people are more vulnerable to sepsis:

- Children < 1 year old and particularly those under 3 months old.
- Children and young people who have had recent (within the last 6 weeks) trauma or surgery or invasive procedures.
- Children with impaired immunity due to chronic illness or drugs (such as steroids, chemotherapy or immunosuppressants).
- Children and young people with indwelling lines / catheters / any breach of skin integrity (such as a burn, blister, cut or skin infection).
- Infants born to mothers who are positive for group B streptococcus during pregnancy or are febrile / unwell during delivery.
- Young women who are pregnant or have been pregnant within the past 6 weeks.

2.3. Assessment:

- 2.3.1. A diagnosis of sepsis should be made on clinical suspicion from predisposing/ risk factors, history, and clinical examination (including observation values compared to age related norms (appendix's 4,5 and 6).
- 2.3.2. The sepsis pathway in this guideline can be used to support this decision making and guide treatment.
- 2.3.3. Children on the Paediatric Assessment Unit (PAU) or the wards who score a 3 or more on the Paediatric Early Warning Score (PEWS)

should have a senior (Tier 2 or 3) medical review within 15 minutes. A decision as to whether they have signs or evidence of sepsis needs to be made and documented either in the notes or on the 'Sepsis pathway'.

- 2.3.4. If sepsis is not suspected and there is no clinical cause for concern or risk factors then use clinical judgement to treat the person, using NICE guidance relevant to their diagnosis when available.

2.4. Divergence from NICE guidance:

- This guidance is based on the updated NICE guidance on sepsis [NG51] from July 2016 and RCHT's tertiary referral centre, Bristol Children's Hospital clinical guideline.
- The algorithm and care pathway have been adapted in an attempt to give an implementable approach to identifying septic children without placing all children with an isolated elevated heart or respiratory rate (due to screaming / agitation / fear for example) in the high risk category.
- It also introduces the concept of correction of heart rate for fever (2 studies^{2,3}), and allows clinical discretion regarding giving fluids if the lactate is between 2 – 4 mmol/L. This decision was based on fact that the NICE guideline development group (NGDG) considered that the current evidence was **not** strong enough to justify determining a particular lactate threshold on a rule-in / rule-out basis for sepsis (based on the low quality of the evidence available).

3. Monitoring compliance and effectiveness

Information Category	Detail of process and methodology for monitoring compliance
Element to be monitored	Compliance with policy/ key changes to practice.
Lead	Consultant Paediatrician (Sepsis Lead).
Tool	Children’s Sepsis six audit tool.
Frequency	Live, on-going audit.
Reporting arrangements	Specialty and Care Group meetings. Audit lead. Paediatric consultants. Paediatric anaesthetists. Emergency Department paediatric lead consultants. Trust lead nurse sepsis.
Acting on recommendations and Lead(s)	Consultant paediatrician (sepsis lead). Trust lead nurse sepsis. Emergency department paediatric lead consultant.
Change in practice and lessons to be shared	Required changes to practice will be identified and actioned within 3 months. A lead member of the team will be identified to take each change forward where appropriate. Lessons will be shared with all the relevant stakeholders.

4. Equality and Diversity

- 4.1. This document complies with the Royal Cornwall Hospitals NHS Trust service Equality and Diversity statement which can be found in the [Equality Diversity And Inclusion Policy](#) or the [Equality and Diversity website](#).
- 4.2. Equality Impact Assessment

The Initial Equality Impact Assessment Screening Form is at Appendix 2.

Appendix 1. Governance Information

Information Category	Detailed Information
Document Title:	Sepsis and Suspected Sepsis- Children (Presenting to Child Health and the Emergency Department) Management Guideline and Care Pathway V2.0
This document replaces (exact title of previous version):	Sepsis and Suspected Sepsis –Children (Presenting to Child Health and the Emergency Department) Management Guideline and Care Pathway V1.0
Date Issued/Approved:	January 2024
Date Valid From:	January 2024
Date Valid To:	January 2027
Directorate / Department responsible (author/owner):	Dr Simon Robertson and Dr Tom Fontaine, Consultant Paediatricians, Child Health
Contact details:	01872 252802
Brief summary of contents:	Clinical Guideline for the management of sepsis in children and young people. Includes clear pathway of care.
Suggested Keywords:	Children. Young people. Paediatric. Sepsis.
Target Audience:	RCHT: Yes CFT: No CIOB ICB: No
Executive Director responsible for Policy:	Chief Medical Officer
Approval route for consultation and ratification:	Child Health Audit and Guidelines Meeting Emergency Department Paediatric Consultants
Manager confirming approval processes:	Caroline Chappell
Name of Governance Lead confirming consultation and ratification:	Tamara Thirlby
Links to key external standards:	Bristol Children's Hospital:
Related Documents:	NICE guideline (NG51) Sepsis: recognition, diagnosis and early management. Overview

Information Category	Detailed Information
	Sepsis: recognition, diagnosis and early management Guidance NICE Hanna CM, Greenes DS. How much tachycardia in infants can be attributed to fever? Ann Emerg Med. 2004 Jun;43(6):699-705. Davies P, Maconochie I. The relationship between body temperature, heart rate and respiratory rate in children. Emerg Med J. 2009 Sep;26(9):641-3.
Training Need Identified?	Yes- Paediatric and Emergency Department Junior Doctors and Nursing staff- familiarity with document.
Publication Location (refer to Policy on Policies – Approvals and Ratification):	Internet and Intranet
Document Library Folder/Sub Folder:	Clinical/ Paediatrics/ Sepsis

Version Control Table

Date	Version Number	Summary of Changes	Changes Made by
December 2019	V1.0	Initial version	Dr Simon Robertson, Consultant Paediatrician
September 2023	V2.0	Full review and revision of document.	Dr Tom Fontaine, Consultant Paediatrician

All or part of this document can be released under the Freedom of Information Act 2000.

All Policies, Strategies and Operating Procedures, including Business Plans, are to be kept for the lifetime of the organisation plus 6 years.

This document is only valid on the day of printing.

Controlled Document.

This document has been created following the Royal Cornwall Hospitals NHS Trust [The Policy on Policies \(Development and Management of Knowledge Procedural and Web Documents Policy\)](#). It should not be altered in any way without the express permission of the author or their Line Manager.

Appendix 2. Equality Impact Assessment

Section 1: Equality Impact Assessment (EIA) Form

The EIA process allows the Trust to identify where a policy or service may have a negative impact on an individual or particular group of people.

For guidance please refer to the Equality Impact Assessment Policy (available from the document library) or contact the Equality, Diversity, and Inclusion Team rcht.inclusion@nhs.net

Information Category	Detailed Information
Name of the strategy / policy / proposal / service function to be assessed:	Sepsis and Suspected Sepsis- Children (Presenting to Child Health and the Emergency Department) Management Guideline and Care Pathway V2.0
Directorate and service area:	Child Health
Is this a new or existing Policy?	Existing
Name of individual completing EIA (Should be completed by an individual with a good understanding of the Service/Policy):	Child Health guidelines group
Contact details:	01872 252802

Information Category	Detailed Information
1. Policy Aim - Who is the Policy aimed at? (The Policy is the Strategy, Policy, Proposal or Service Change to be assessed)	Clear guidance for the treatment of children and young people with sepsis.
2. Policy Objectives	To provide clear guidance for the treatment of children and young people with sepsis.
3. Policy Intended Outcomes	Evidence based standardised practice.
4. How will you measure each outcome?	As per Trust Sepsis audit.
5. Who is intended to benefit from the policy?	Staff and patients.
6a. Who did you consult with? (Please select Yes or No for each category)	- Workforce: - Patients/ visitors: - Local groups/ system partners: - External organisations: - Other: Yes No No No No

Information Category	Detailed Information
6b. Please list the individuals/groups who have been consulted about this policy.	Please record specific names of individuals/ groups: Child Health Audit and Guidelines Group. Emergency Department Paediatric Consultants. Medicines Practice Committee.
6c. What was the outcome of the consultation?	Approved- 16 November 2023.
6d. Have you used any of the following to assist your assessment?	National or local statistics, audits, activity reports, process maps, complaints, staff, or patient surveys: No.

7. The Impact

Following consultation with key groups, has a negative impact been identified for any protected characteristic? Please note that a rationale is required for each one.

Where a negative impact is identified without rationale, the key groups will need to be consulted again.

Protected Characteristic	(Yes or No)	Rationale
Age	No	
Sex (male or female)	No	
Gender reassignment (Transgender, non-binary, gender fluid etc.)	No	
Race	No	Any information provided should be in an accessible format for the parent/ carer/ patient's needs- i.e., available in different languages if required/ access to an interpreter if required.
Disability (e.g. physical or cognitive impairment, mental health, long term conditions etc.)	No	Those parent/ carer/ patients with any identified additional needs will be referred for additional support as appropriate- i.e., to the Liaison team or for specialised equipment. Written information will be provided in a format to meet the family's needs e.g., easy read, audio etc.
Religion or belief	No	All staff should be aware of any beliefs that may impact on the decision to treat.
Marriage and civil partnership	No	All staff should be aware of any marital arrangements that may have an impact on care (for example: separated parents, domestic abuse).

Protected Characteristic	(Yes or No)	Rationale
Pregnancy and maternity	No	
Sexual orientation (e.g. gay, straight, bisexual, lesbian etc.)	No	

A robust rationale must be in place for all protected characteristics. If a negative impact has been identified, please complete section 2. If no negative impact has been identified and if this is not a major service change, you can end the assessment here.

I am confident that section 2 of this EIA does not need completing as there are no highlighted risks of negative impact occurring because of this policy.

Name of person confirming result of initial impact assessment: Child Health Audit and Guidelines Group.

If a negative impact has been identified above OR this is a major service change, you will need to complete section 2 of the EIA form available here:
[Section 2. Full Equality Analysis](#)

Appendix 3. Paediatric Sepsis Screening and Action Pathway

Paediatric Sepsis Screening and Action Pathway

Date:
Time:
Location:

Patient ID sticker

RECOGNISE

PEWS ≥ 3 or concern child unwell?
Call for Tier 2 or 3 review within 15mins and complete risk assessment

Are 2 (or more) of below present?:	Yes/No	Value
Temp $<36^{\circ}$ or $>38.5^{\circ}$ ($>38^{\circ}$ in oncology patients)		
Tachycardia (triggering PEWS?)		
Tachypnoea (triggering PEWS?)		
Altered mental state? (sleepy/floppy/lethargic/irritable)		
Mottled or cap refill ≥ 3 secs		
Spontaneously report pain in legs		
OR any 1 RED FLAG		
Temp $\geq 38^{\circ}$ in a child ≤ 3 months		
Hypotension (see definition below)		
Feel and record pulse volume/character		
Bradycardia (<60 bpm)		
Lactate >4 mmol/l (if $>2 < 4$ and any doubt consider a red flag)		
SpO2 $<90\%$ /grunting /apnoea		
Anuria last 18hours/UO <0.5 mls/kg/hr		
Immunocompromised		
Non-blanching rash		
CLINICALLY V.UNWELL/SEPTIC		

IF NO/decision made NOT SEPTIC by Tier 2 or 3 Clinician

YES

START SEPSIS PATHWAY

Call for immediate Tier 2 or 3 attendance

CONSIDER 2222 call

Tier 2/3 clinician informed (name):

By (name):

Date/time:

Not sepsis

Document assessment, diagnosis and plan below, exit pathway continue appropriate management.

Unsure?

Document assessment here, cont. 15 min obs and then review in 1hr (page 4)

Take FBC, CRP, U+E, + qas

Name:

Date/time:

Signature:

Role:

Unsure? Pg. 4

RESPOND

Paediatric Sepsis 6: Within 1 hour

	Time	Sign
1 High flow oxygen (Document below with reason if not required)		
2 Record blood pressure, start urine collection and fluid balance (Fresh nappy, fluid balance chart)		
3 Obtain IV/IO access (repeat IV attempts must not delay antibiotics)		
4 Take blood cultures, blood gas (As priority include FBC, U+E, LFT, Coag, PCR)		
5 Consider IV/IO fluids, aim for normal physiological parameters (If haemodynamic compromise give 10ml/kg of 0.9% saline or Plasmolyte and prep x3 more)		
6 ≤ 1 month OR < 41 wks CGA - IV CefOTaxIME(50mg/kg) + Amoxicillin(50mg/kg) 1-3 months - IV CeftRaxone(80mg/kg) + Amoxicillin(50mg/kg) ≥ 3 months - IV CeftRaxone(80mg/kg) Think: If neutropenic, neonate (<72 hrs) or have vascular line – check antimicrobial policy		

CONTINUE SEPSIS ASSESSMENT ON PAGE 2

Document any reason for deviation from above/other comments, specifically, if no O₂ or fluids given or delay in abx document why.

HYPOTENSION

is systolic $<5^{\text{th}}$ centile: <1 year <70 mmHg, 1-3yrs <80 mmHg, 3-12yrs <90 mmHg

In an otherwise stable child heart rate can be corrected for fever by deducting 10bpm per degree $>37^{\circ}$ with caution and documentation of decision.

Role:

Date/Time:

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Name:

DOB

CRN

Patient ID sticker here:

Name:

Grade:

Date/Time(of your assessment):

Always think: DO YOU NEED HELP?

PRINT SEPSIS PROTOCOL +/- PICU DRUG SHEET

Clinical Assessment

A+B: Any support required?

C: Do you need to prepare fluids and Inotropes?

D: Do you need to consider Aciclovir? Is Glucose OK? Any concern of NAI?

E: Any rash/bleeding/ sign of injury? Abdo OK?

F: Number of boluses? What fluid? Is maintenance running? Any UO? Consider catheter – aim 0.5mls/kg/hr

Impression:

- Plan:
- **Senior medical review within 1 hour = (time)**
 - **Antibiotics** (which and time given).....
 - **IV fluids:** (type and rate).....
 - **Fluid balance/UO** (mls/kg/hr).....
 - **Observations every:** 5 mins or 15 mins (circle which)
 - **Repeat gas in max. 1 hour = (time)**.....
 - **Check formal bloods** (FBC, clotting, electrolytes Mg2+, renal)
 - **Other:**

nO₂:

Sats:

RR:

HR:

CRT:

BP:

Venous/Cap/Art?

Sample time:

pH:

BE:

HCO₃:

Lactate:

CO₂:

O₂:

SaO₂:

Glucose:

STOP!

Check 'Sepsis 6' on page one complete and times/reasons for any deviation recorded. (tick) ☐

Signature:

Role:

Print:

Date/Time:

Name:

DOB

CRN

Patient ID sticker here:

Name + Grade (Tier2/3):

Date/Time (of assessment):

REASSESS

WITHIN 1 HOUR of starting treatment

Yes/No/Value

Date/Time

1 HR and/or RR remain above age specific normal range or CRT >3 seconds or Hypotensive

2 Venous/Arterial Lactate >2mmol/l (If >4 call ITU)

3 Signs of fluid overload/Heart failure?
(Wheeze/crackles/liver edge)

**If YES to ANY of the above escalate – Consultant, ITU, WATCH (tel: 03000300789)
CONSIDER 2222**

Who called:

Time called:

Response:

- Is the patient responding to treatment?
- Is the patient shocked? End organ dysfunction, lactate, base deficit increasing?
- Do you need to get help? 2222/ITU/WATCH
- Do you need inotropes, blood products?
- Is the diagnosis correct? DKA/adrenal insufficiency/metabolic, malignancy, cardiac, trauma/NAI

PRINT PICU DRUG SHEET

A+B: Crackles/new wheeze? Any support required?

C: As 2nd saline 20mls/kg bolus given, prepare inotropes (dopamine, adrenaline or dobutamine). Is liver edge palpable?

D: Do you need to consider Acidosis? Is Glucose OK? Any concern of NAI?

E: Any rash/bleeding/trauma/Injury? Abdo OK?

F: Number of boluses? What fluid? Is maintenance running? UO? Consider catheter – aim 0.5mls/kg/hr

IMPRESSION:

PLAN: Next clinical review time:

Likely disposition:

Ward / HDU / Critical Care (is relevant team aware?)

FiO₂:

Sats:

RR:

HR:

CRT:

BP:

Venous/Cap/Art?

Sample time:

pH:

BE:

HCO₃⁻:

Lactate:

CO₂:

SpO₂:

SaO₂:

Glucose:

Ca²⁺:

Signature:

Role:

Print:

Date/Time:

UNLIKELY SEPSIS – (page 4)

Name:

DOB

CRN

Patient ID sticker here:

Review at 1 hour:

Date/Time:

Working Diagnosis:

Assessment: (if lactate >2 or evidence of AKI if >12yrs or increasing clinical concern RX as sepsis)

FBC:

CRP:

Lactate:

BSL:

Creatinine/evidence of AKI:

Decision (circle):

SEPSIS	NOT SEPSIS	UNCLEAR
Start Sepsis 6 (complete new booklet)	Document diagnosis and continue appropriate treatment	15 min obs and review in 1 hour

Diagnosis:

Signature:

Role:

Print:

Date/Time:

Review at 2 hours:

Date/Time:

Working Diagnosis:

Assessment:

Decision (circle):

SEPSIS	NOT SEPSIS	UNCLEAR
Start Sepsis 6	Document diagnosis and continue appropriate treatment	15 min obs and review in 1 hour Consider treating for presumed infection with antibiotics (following blood culture)

Diagnosis and plan:

Signature:

Role:

Print:

Date/Time:

Appendix 4. Risk Stratification Tool for Adults, Children and Young People Aged 12 Years and Over with Suspected Sepsis

[Please refer to table 1 of the NICE Sepsis Risk Stratification Tools](#)

Appendix 5. Risk Stratification Tool for Children Aged 5-11 Years with Suspected Sepsis

[Please refer to table 2 of the NICE Sepsis Risk Stratification Tools](#)

Appendix 6. Risk Stratification Tool for Children Aged Under 5 Years with Suspected Sepsis

[Please refer to table 3 of the NICE Sepsis Risk Stratification Tools](#)